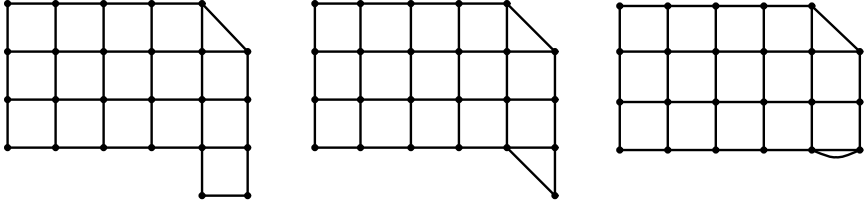
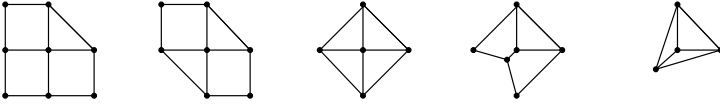


This way we have deleted the first square in the last row; similarly, we can delete the second square in this row, and so on; the last square is deleted by two series reductions and one parallel reduction.



We can delete all the squares in the first column of our board symmetrically, if $n \geq 4$.



At the end, this leaves us with the grid $G(3, 3)$ with one short corner, which is easily reduced first to the “wheel” graph W_4 of a square pyramid, and then to K_4 . $\square$

Corollary 4.7. *Every 3-connected planar graph G can be reduced to K_4 by a sequence of simple ΔY transformations.*

Proof. The three lemmas together show that G is ΔY -reducible. We follow this reduction to the first point where parallel or series edges are created. These can be reduced immediately, which also results in a 3-connected planar graph G by Lemma 4.2. The graph we now have has fewer edges; hence we are done by induction on size. $\square$

Corollary 4.7 also completes our proof of Steinitz' Theorem 4.1.

4.4 Extensions of Steinitz' Theorem

Corollary 4.8. *For every 3-polytope, there is a rational 3-polytope that is combinatorially equivalent.*

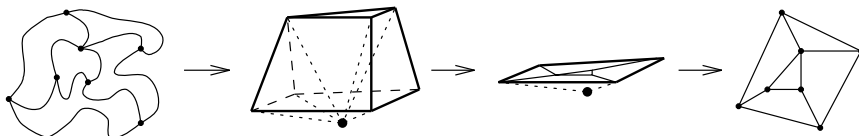
Proof. This follows from our proof of Steinitz' theorem: we have shown in fact that every 3-polytope can be reduced to a simplex by two types of operations $P \longrightarrow P'$:

- constructing a combinatorially polar polytope,
- “cutting off” a vertex of degree 3.

In both cases, if we have a rational realization of P' , then we can also construct P with rational vertices and inequalities. $\square$

Corollary 4.9. *Every 3-connected planar graph has a representation in the plane such that all edges are straight, and all the bounded regions determined by it, as well as the union of all the bounded regions, are convex polygons.*

Proof. This we get by representing G as the graph of a 3-polytope, and then choosing a point $\mathbf{x}$ beyond one of the facets. “Viewing” the polytope from this perspective (and projecting it to the facet along the “rays of vision”) gives us the required representation. $\square$



This proof contains the construction of a “Schlegel diagram” for the 3-polytope: see the next lecture. There are various results in graph theory related to this. There are other proofs, like the one by Tutte [549], which use less geometry and more graph theory. A reduction method by Thomassen [540] leads to a very short proof [268, Anhang 1]. See also Exercise 4.7.

There are various other strengthenings of Steinitz' theorem, which say that we can prescribe a lot about the polytope to be constructed. For example, one can *prescribe the shape* of a facet of P in advance, by Barnette & Grünbaum [52]. That is, given a 3-polytope P with a k -gon facet K , and given any k -gon $K' \subseteq \mathbb{R}^3$, we can “redraw” P so that we get a polytope $P' \subseteq \mathbb{R}^3$ which is combinatorially equivalent to P , and such that K' is the facet of P' which corresponds to $K \subseteq P$.

Similarly, we can *prescribe the shadow boundary*, by Barnette [41]: for every cycle in $G(P)$ we can find a realization of P and a projection of P to the plane that carries the cycle to the boundary of the image polygon. This is quite amazing: just try to verify this on your favorite 3-polytope; see Exercise 4.9. However, one shouldn't get too greedy at this point: the shape of the image polygon *cannot* be prescribed for this — see Exercise 4.12.

Also, for every symmetric graph there is a polytope that realizes the full symmetry group of the graph, see Mani [375] – and Theorem 4.13 below.

The following is perhaps the more important extension, and is what properly should be taken as a part of the Steinitz theorem. It says that the space of all ways to find coordinates for a 3-polytope is connected (modulo reflections). For the stronger statement below, we have to consider all coordinatizations “modulo rotations and reflections.” For this, we need a few definitions.

Definition 4.10. Let P be a d -polytope on $n > d$ vertices. We can label the vertices $\text{vert}(P) = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ such that $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{d+1}\}$

determines a flag, that is, $\text{aff}(\{\mathbf{x}_1, \dots, \mathbf{x}_k\}) \cap P$ is a $(k-1)$ -face of P , for $1 \leq k \leq d+1$.

The realization space

$$\mathcal{R}(P) \subseteq \mathbb{R}^{d \times n}$$

of P is the set of all matrices $Y \in \mathbb{R}^{d \times n}$ such that $\mathbf{y}_k = \mathbf{x}_k$ for $1 \leq k \leq d+1$, and such that P is combinatorially equivalent to $Q := \text{conv}\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ under the correspondence $\mathbf{x}_i \rightarrow \mathbf{y}_i$.

It is easy to see that the realization space is an *elementary semialgebraic set* defined over $\mathbb{Z}$, that is, a subset of a real vector space that can be defined in terms of polynomial equations and strict inequalities with integer coefficients. Such semialgebraic sets can be arbitrarily complicated as topological spaces, in general (see Exercise 4.22). The following assumes that you know what *contractible* means for a topological space: if not, just accept that it says that the space has “no holes”; in particular, contractible spaces are connected.

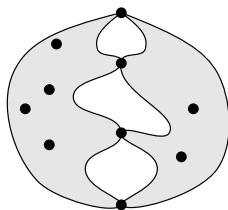
Theorem 4.11 (Steinitz’ theorem). [527, 524]

For every 3-polytope P , the realization space $\mathcal{R}(P)$ is contractible, and thus connected.

All of these theorems — on prescribed facets, shadow boundary, symmetry, realization space, and so on — were proved with basically the same technique, and some clever variations. Like Steinitz’ original theorem, they are far from trivial. One way to see this is that they all fail “one dimension up,” for $d = 4$. We will construct explicit examples in the next two lectures.

Notes

The reason for this version of the definitions for graph connectivity is that they fit into a larger and very natural pattern. Namely, following Tutte [550, 551] (see Truemper [547, p. 15]) one defines a *k-separation* (for $k \geq 1$) of a graph $G = (V, E)$ as a partition into two graphs, $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, which have exactly k vertices but no edges in common, and which have at least k edges each. Thus, our sketch



shows a 4-separation, if each side has at least 4 edges (and thus at least one cycle, or a nonseparating vertex, or both). For any $k \geq 2$, a connected graph is *k-connected* if it has no *l*-separation for $1 \leq l < k$.